

L09&15_Q03: Compute the numerical values of the approximate solution of the following IVP at $x=$

$$0:0.2:2, y'(x) = y - x^2 + 1, y(0) = 0.5$$

%Compute the numerical values of the approximate solution of the following IVP

% at $x= 0:0.2:2, y'(x)=y-x^2+1, y(0)=0.5$

clc

clear all

close all

f=@(x,y) (y-x.^2+1)

[x, ya] =ode45(f, 0:0.2:2, 0.5)

plot(x, ya)

L09&15_Q04: Find the exact solution of the following IVPs, then compute its approximate solution at $x=0:0.2:2$ using ode45 command. Finally, makes the table of values at t,ya and y(exact solution).

$$y'(x) = y - x^2 + 1, y(0) = 0.5$$

%Find the exact solution of the following IVPs,

%then compute its approximate solution at $x=0:0.2:2$ using ode45 command.

%Finally, makes the table of values at t,ya and y(exact solution).

% $y'(x)=y-x^2+1, y(0)=0.5$

clc

clear all

close all

eq='Dy=y-x^2+1';

initialcond='y(0)=0.5';

y=dsolve(eq,initialcond,'x')

f=@(x,y) (y-x.^2+1)

[x, ya] =ode45(f, 0:0.2:2, 0.5)

[x, ya, 2.*x-exp(x) ./2+x.^2+1]

L09_Q09: Compare the exact and approximate solution of $\frac{dy}{dt} = t + y$, $y(0) = 3$, $tspan = [0,5]$ by using ode45 command.

```
%Compare the exact and approximate solution of dy/dt=t+y ,  
%y(0)=3,tspan=[ 0,5] by using ode45 command  
clc  
clear all  
close all  
format short  
f=@(t,y) (t+y);  
tspan=[ 0,5];  
y0=3;  
[ t,y]=ode45(f,tspan,y0);  
exactsoln=dsolve('Dy=t+y','y(0)=3','t');  
exact=eval(vectorize(exactsoln));  
table=[ t,y,exact]  
plot(t,y,'b',t,exact,'r')  
legend('Approximate','Exact')  
xlabel('x-axis')  
ylabel('y-axis')  
title('Comparing exact and approximate solution of ODE')
```

L10_Q15: Generate a diagonal matrix by using a vector and diag built-in function.

```
% Generate a diagonal matrix by using a vector and diag built-in function.  
clc  
clear all  
close all  
echo on  
A=[ 1,2,3,4]  
B=diag(A)  
echo off
```

L10_Q20: Find determinant of a matrix.

```
% Find determinant of a matrix.
clc
clear all
close all
echo on
A=[ 1 2 3;4 5 6;5 2 1];
B=det(A)
echo off
```

L10_Q21: Find inverse matrix of a matrix.

```
% Find inverse matrix of a matrix.
clc
clear all
close all
format short
echo on
A=[ 1 2 3;4 5 6;5 2 1];
B=inv(A)
echo off
```

L10_Q22: Multiplication of two matrixes by using for loop.



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```
% Multiplication of two matrixes by using for loop.
clc
clear all
close all
format short
echo on
A=[ 1 2 3;4 5 6];
%order of A=2 by 3
B=[ 3 4 6;4 5 1;4 3 1];
%order of B=3 by 3
for i=1:2
    for j=1:3
        C(i,j)=A(i,:)*B(:,j)
    end
end
C
C1=A*B
echo off
```

L11_Q29: Compute the Eigen values and Eigen vectors of the following matrix

$$B = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 4 & 1 \\ 6 & 1 & 8 \end{bmatrix}$$

```
% Compute the Eigen values and Eigen vectors of the following matrix
clc
clear all
close all
echo on
A=[ 0,1,3;2,4,1;6,1,8]
eig(A)
[ V,D]=eig(A)
echo off
```

L11_Q32: Solve following system of linear equations by using **solve** command.

$$3x + 2y + z = 1, \quad y = 2, \quad x + 2y = 3$$

```
% Solve following system of linear equations by using solve command.  
%3x+2y+z=1; y=2; x+2y=3  
clc  
clear all  
close all  
echo on  
eq1='3*x+2*y+z=1'  
eq2='y=2'  
eq3='x+2*y=3'  
soln=solve(eq1,eq2,eq3)  
X=double([ soln.x,soln.y,soln.z] )'  
echo off
```

L11_Q34: Consider the system of linear equations;

$$3x + 2y + z = 1, \quad y = 2, \quad x + 2y = 3$$

Enter it into MATLAB as an augmented matrix named Ab; Use elementary row operations to reduced echelon form.

```
% Consider the system of linear equations;  
% 3x+2y+z=1; y=2 ;    x+ 2y=3  
% Enter it into MATLAB as an augmented matrix named Ab;  
% Use elementary row operations to reduced echelon form.  
clc  
close all  
clear all  
echo on  
format short  
A=[ 3,2,1;0,1,0;1,2,0] ;  
b=[ 1,2,3] ' ;  
Ab=[ A,b] ;  
Ab(1,:)=1/3*Ab(1,:);  
Ab(3,:)=Ab(3,:)-Ab(1,:);  
Ab(1,:)=Ab(1,:)-0.6667*Ab(2,:);  
Ab(3,:)=Ab(3,:)-1.3333*Ab(2,:);  
Ab(3,:)=1/-0.3333*Ab(3,:);  
Ab(1,:)=Ab(1,:)-0.3333*Ab(3,:);  
X=Ab(:,4)
```


L12&15_Q43: For the integral $\int_1^5 \sqrt{1+x^3} dx$ compute the Riemann sum approximation using 100 equally spaced sub-intervals and left end points.

```
% For the integral sqrt((1+x^3))dx [ 1, 5]
% compute the Riemann sum approximation
% using 100 equally spaced sub-intervals and left end points.
function value=riemannsum(f,a,b,n)
clear all;
close all;
clc;
a=1;
b=5;
n=100;
value=0;
f=@(x) (sqrt(1+x.^3));
dx=(b-a)/n;
for k=1:n
    c=a+(k-1)*dx;
    value=value+f(c);
end
value=dx*value;
```


L12&15_Q48: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Trapezoid Rule considering 10 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [ 1, 5]
% using Trapezoid Rule considering 10 equally spaced sub-intervals.
close all
clear all
clc
a=1;
b=5;
n=10;
dx=(b-a)/n;
f=@(x)(sqrt(1+x.^3));
AW=(f(a)+f(b))/2;
for k=1:n-1
    x(k)=a+k*dx;
end
funvalue=f(x);
answer=(sum(funvalue)+AW)*dx
```

L12&15_Q54: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's One-Third Rule considering 300 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [ 1, 5]
%Simpson's one Third Rule by considering 300 equally spaced sub-intervals.
close all
clear all
clc
a=1;
b=5;
n=300;
dx=(b-a)/n
f=@(x) (sqrt(1+x.^3));
value1=0;
value2=0;
for k=1:n-1
    x(k)=a+k*dx;
    if mod(k,2)==0
        value1=value1+2*f(x(k));
    else
        value2=value2+4*f(x(k));
    end
end
answer=(value1+value2+f(a)+f(b))*(dx/3)
```


L12&15_Q58: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's Three-Eight Rule considering 30 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [ 1, 5]
% using Simpson's Three-Eight Rule by considering 30 equally spaced sub-intervals.
clear all;
clc;
close all;
sum1=0;
sum2=0;
a=1;
b=5;
n=30;
f=@(x) sqrt(1+x^3)
dx=(b-a)/n;
value=f(b)+f(a);
for k=1:n-1
    c=a+k*dx;
    if mod(k,3)==0
        sum1=sum1+2*f(c);
    else
        sum2=sum2+3*f(c);
    end
end
sum=sum1+sum2;
Answer=(value+sum)*(3*dx/8)
```

L12&15_Q61: Compute the Riemann sum of the function $f(x) = e^{-x^2}$ for $n=4, 6, 8, 10, 100$ and 1000 by choosing C_k as the right end point of the interval $[0, 2]$.

```
% Compute the Riemann sum of the following function f(x) = e^-x^2 for
% n=4,6,8,10,100,1000 by choosing ck as the right point of interval[0,2]
clear all
close all
clc
a=0;
b=2;
n=4;
%ALSO CHECK RESULTS BY PUTTING n=6, n=8, n=10, n=100 and n=1000
dx=(b-a)/n
f=@(x)exp(-x.^2)
for k=1:n
    c(k)=a+k*dx;
end
funcvalue=f(c);
summation=sum(funcvalue);
answer=summation*dx
```


L13_Q64: Construct the function vector $p_n^{(\alpha,\beta)}(x)$ (Orthogonal Shifted Jacobi Polynomials) defined on $[0,1]$ by $\alpha = 1$ and $\beta = 2$ and scale level $M=10$ taking using equation

$$p_n^{(\alpha,\beta)}(x) = \sum_{k=0}^n \frac{(-1)^{(i-k)} \Gamma(i + \beta + 1) \Gamma(i + k + \alpha + \beta + 1)}{\Gamma(k + \beta + 1) \Gamma(i + \alpha + \beta + 1) \Gamma(i - k + 1) \Gamma(k + 1) T^k} x^k \text{ where } i = 0, 1, 2, 3, \dots$$

% Construct the function vector (Orthogonal Shifted Jacobi Polynomials)
% defined on $[0,1]$ by alpha=1 and beta=2 and scale level M=10 taking using equation

```
function P=JacobFunVec(alpha,beta,M)
```

```
clear all
```

```
close all
```

```
clc
```

```
syms x T k
```

```
M=10;
```

```
T=1;
```

```
alpha=1;
```

```
beta=2;
```

```
for i=0:M-1
```

```
    a1=(-1)^(i-k);
```

```
    a2=gamma(i+beta+1);
```

```
    a3=gamma(i+k+alpha+beta+1);
```

```
    d1=gamma(k+beta+1);
```

```
    d2=gamma(i+alpha+beta+1);
```

```
    d3=gamma(i-k+1);
```

```
    d4=gamma(k+1);
```

```
    d5=T^k;
```

```
    aa=(a1*a2*a3)/(d1*d2*d3*d4*d5);
```

```
    P(i+1)=symsum(aa*x^k,k,0,i);
```

```
end
```

```
P=transpose(P);
```


L13_Q71: Compute the constant factor defined in $R_{T,j}^{(\alpha,\beta)} \delta_{(i,j)} = \frac{T^{\alpha+\beta+1} \Gamma(j+\alpha+1) \Gamma(j+\beta+1)}{(2j+\alpha+\beta+1) \Gamma(j+1) \Gamma(j+\alpha+\beta+1)}$ for choosing the scale level $M=7$ and $\alpha = 3$ and $\beta = 1$.

% Compute the constant factor defined for choosing the scale level $M=7$ and $\alpha=3$ and $\beta=1$.

```
function R=ConstantFactor(alpha,beta,T,M)
```

```
clear all
```

```
close all
```

```
clc
```

```
syms j alpha beta T
```

```
T=1;
```

```
alpha=3;
```

```
beta=1;
```

```
M=7;
```

```
for j=0:M-1
```

```
    a1=T^(alpha+beta+1);
```

```
    a2=gamma(j+alpha+1);
```

```
    a3=gamma(j+beta+1);
```

```
    d1=(2*j+alpha+beta+1);
```

```
    d2=gamma(j+1);
```

```
    d3=gamma(j+alpha+beta+1);
```

```
    R(j+1)=(a1*a2*a3)/(d1*d2*d3);
```

```
end
```

```
R=transpose(R)
```


L13&15_Q74: Compute the series coefficients C_i of Jacobi polynomials for scale level $M=6$ and $\alpha = 2$ and $\beta = 1$. While

$$C_i = \frac{1}{R_{T,j}^{(\alpha,\beta)}} \int_0^T f(x) W_T^{(\alpha,\beta)}(x) P_i^{(\alpha,\beta)}(x) dx$$

% Compute the series coefficients C_i of Jacobi polynomials
% for scale level $M=6$ and $\alpha=2$ and $\beta=1$.

function C=JacobiCoeffVector(f,x,M,alpha,beta,T)

close all;

clear all;

clc;

syms f x M alpha beta T k i j

T=1;

alpha=2;

beta=1;

M=6;

W = (T - x).^alpha * x^beta;

for i=0:M-1

 a1 = (-1)^(i-k);

 a2 = gamma(i + beta+1);

 a3 = gamma(i + k+ alpha + beta +1);

 d1 = gamma(k+beta +1);

 d2 = gamma(i + alpha + beta + 1);

 d3 = gamma(i - k + 1);

 d4 = gamma(k+1);

 d5 = T^k;

 aa = (a1*a2*a3)/(d1*d2*d3*d4*d5);

 P(i+1) = symsum(aa*x^k, k, 0, i);

end

P;

PW=f*P*W;

for j = 0:M-1

 a1 = T^(alpha + beta +1);

 a2 = gamma(j + alpha + 1);

 a3 = gamma(j + beta + 1);

 d1 = 2*j + alpha + beta + 1;

 d2 = gamma(j + 1);

 d3 = gamma(j + alpha + beta + 1);

 R(j + 1) = (a1*a2*a3)/(d1*d2*d3);

end

R ;

RR=1./R;

for i = 0:M-1

 C(i+1)=int(PW(i+1),x,0,i)*RR(i+1);

end

C=transpose(C);

L14_Q77: Write MATLAB code to compute the Legendre Polynomials using the relation $P_0(z) = 1$, $P_1(z) = z$, $P_2(z) = \frac{1}{2}(3z^2 - 1)$, $P_3(z) = \frac{1}{2}(5z^3 - 3z)$ and $P_4(z) = \frac{1}{8}(35z^4 - 30z^2 + 3)$ at $z=0$, $z=1$ and $z=5$ and using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1}zL_i(Z) - \frac{i}{i+1}L_{i-1}(Z)$.

% Write MATLAB code to compute the Legendre Polynomials using the relation
% for $i=1:20$ and at $z=0$, $z=1$ and $z=5$

```
function p=LegenderFunctionVector(M,T,value1,value2)
clear all;
close all;
clc;
syms z
z=0;
%ALSO FIND RESULTS BY PUTTING Z=1 AND Z=5
format short
M=20;
value(1)=z;
value(2)=sym(3/2)*z^2-sym(1/2);
for i=1:M
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);
end
p=transpose(value);
```


L14&15_Q82: Construct the function vector $P_n(x)$ (Orthogonal Shifted Legendre Polynomials) defined on $[0,1]$ for the scale level $M=30$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)(\Gamma(k+1))^2}$ where $i = 0,1,2,3,\dots$. Moreover compute the values at of $P_n(x)$ at $x=0$.

```
% Construct the function vector Orthogonal Shifted Legendre Polynomials  
% defined on [0,1] for scale level M=30 using the relation at x=0;
```

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)
```

```
clear all;
```

```
close all;
```

```
clc;
```

```
syms x T k i;
```

```
M=30;
```

```
T=1;
```

```
x=0;
```

```
for i=0:M-1
```

```
    a1=(-1)^(i+k);
```

```
    a2=gamma(i+k+1);
```

```
    d1=gamma(i-k+1);
```

```
    d2=(gamma(k+1))^2;
```

```
    aa=(a1*a2)/(d1*d2);
```

```
    p(i+1)=symsum(aa*x^k,k,0,i);
```

```
end
```

```
p=transpose(p);
```

L14&15_Q87: Compute the constant factor defined in R_j for choosing scale level $M=40$ and using the relation $R_j = \frac{1}{2j+1}$

% Write MATLAB code for generating the constant factor
% for choosing scale level $M=40$ and using the relation $R_j=1/(2j+1)$

```
function p=ConstantFactor(j,M)
```

```
close all;
```

```
clear all;
```

```
clc;
```

```
M=40;
```

```
format short;
```

```
syms j
```

```
for j=0:M-1
```

```
    a1=1;
```

```
    a2=2*j+1;
```

```
    R(j+1)=a1/a2;
```

```
end
```

```
R
```


L14_Q90: Compute the series coefficients C_i for choosing scale level $M=10$ and using the relation

$$C_i = \frac{1}{R_i} \int_0^1 f(x) P_i(x) dx$$

%Write MATLAB code to compute the series coefficients for scale level $M=10$.

```
function P=Lagdrfunvec(M,T)
clc;
clear all;
close all;
syms x T k
M=10;
for i=0:M-1
    a1=(-1)^(i+k);
    a2=gamma(i+k+1);
    d1=gamma(i-k+1);
    d2=(gamma(k+1))^2;
    aa=(a1*a2)/(d1*d2);
    P(i+1)=symsum(aa*x^k,k,0,i);
end
P=transpose(P);
function R=Lagendrorthogonal(i,M)
syms i
M=10;
for i=0:M-1
    a1=1;
    a2=2*i+1;
    R(i+1)=a1/a2;
end
R
function C=Lagendrcoefvec(i,T,f)
syms x T i f
P=LagdrFunVec(T);
R=LagdrOrthogonal(i);
M=10;
Kw=f*P;
for i=0:M
    aa=Kw(i+1);
    C(i+1)=(1/(R(i+1)))*int(aa,x,0,T);
end
C=transpose(C);
```

L14_Q92: Compute the values of the Legendre Polynomials using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1}zL_i(Z) - \frac{i}{i+1}L_{i-1}(Z)$ at $z = 3, z = 4$ and $z = 6$

% Write MATLAB code to compute the Legendre Polynomials using the relation
% at $z=3, z=4$ and $z=6$

```
function p=LegenderFunctionVector(M,T,value1,value2)
```

```
clear all;
```

```
close all;
```

```
clc;
```

```
syms z
```

```
z=3;
```

```
%ALSO FIND RESULTS BY PUTTING Z=4 AND Z=6
```

```
format short
```

```
M=10;
```

```
value(1)=z;
```

```
value(2)=sym(3/2)*z^2-sym(1/2);
```

```
for i=1:M
```

```
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);
```

```
end
```

```
p=transpose(value);
```


L15_Q93: Construct the one-dimensional Orthogonal Shifted Legendre function vector $P_i(x)$ defined on $[0,1]$ for $i=0:20$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)(\Gamma(k+1))^2}$ where $i = 0,1,2,3, \dots$

% Construct the function vector Orthogonal Shifted Legendre Polynomials
% defined on $[0,1]$ for $i=0:20$ using the relation;

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)
```

```
clear all;
```

```
close all;
```

```
clc;
```

```
syms x T k i;
```

```
T=1;
```

```
M=20;
```

```
for i=0:M
```

```
    a1=(-1)^(i+k);
```

```
    a2=gamma(i+k+1);
```

```
    d1=gamma(i-k+1);
```

```
    d2=(gamma(k+1))^2;
```

```
    aa=(a1*a2)/(d1*d2);
```

```
    p(i+1)=symsum(aa*x^k,k,0,i);
```

```
end
```

```
p=transpose(p);
```

L15_Q96: Compute the following relation for $i=0:10$ and choosing the values of $c=1$ and $d=2$:

$$\Omega_i^{(c,d)}(x) = \frac{(4)^{c+d+10} \Gamma(i+5c+d+10)(i+19)!}{(2i+c+d+12)\Gamma(i+c+5)(i+10c+d+15)!}$$

% Compute the following relation for $i=0:10$ and choosing the values of $c=1$ and $d=2$:

```
function P=JacobFunVec(c,d,i)
clear all;
close all;
clc;
syms x k
n=10;
c=1;
d=2;
for i=0:n
    a1=(4)^(c+d+10);
    a2=gamma(i+5*c+d+10);
    a3=factorial(i+19);
    d1=2*i+c+d+12;
    d2=gamma(i+c+5);
    d3=factorial(i+10*c+d+15);
    aa=(a1*a2*a3)/(d1*d2*d3);
    P(i+1)=aa;
end
P=transpose(P);
```


Q3:- $A = \begin{bmatrix} 2 & 3 & 4 & 7 \\ 3 & 2 & 1 & 8 \\ 7 & 8 & 9 & 10 \\ 4 & 5 & 6 & 2 \end{bmatrix}$

- Determine square matrix whose columns represent the Eigenvectors of the matrix A .
- Determine the diagonal matrix having the Eigenvalues of the matrix A on its main diagonal.